Family Name _____

First Name _____

Student Number |__|__|__|__|__|__|__|__|

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UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Computer Science and Software Engineering

EXAMINATION

End-of-year Examinations, 2022

ENCE260-22S2 (C) Computer Systems

Examination Duration: 120 minutes

Exam Conditions:

Closed Book exam: Students may not bring in any written or printed materials.

Calculators with a 'Civil' or 'UC' sticker approved.

Materials Permitted in the Exam Venue:

None

Materials to be Supplied to Students:

1 x Write-on question paper/answer book

Instructions to Students:

ALL questions are COMPULSORY.

There are FOUR questions in total.

Maximum TOTAL marks 100.

For Examiner Use Only

Question	Mark

Total _____

INSTRUCTIONS

You are required to complete this examination by writing your answers into this booklet. Show your working as you complete this examination booklet, so that you can more easily receive partial credit.

If you need more space for your answers, several extra pages can be found at the end of this booklet. Make a note indicating when your work is continued on an extra page.

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Exam starts on Page 4

Q1. **(a)** You are stuck on a desert island, but thankfully a plane flies overhead! It drops a box of supplies, in which is a box of assorted fundamental logic gates, a radio that has three digital outputs A, B, and C, and one input S, and a note that reads : Implement the following truth table if you need to be rescued: $S = f(A, B, C)$.

A	B	C	S
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(i) **[2 marks]**

Write a Boolean expression for S in "sum-of-minterms" form (i.e., each term in the expression should contain each of the input symbols).

(ii) **[3 marks]**

Using the rules of Boolean algebra, show how the "sum-of-minterms" expression for S can be simplified to obtain the Boolean expression

$$S = \overline{A + B} + BC.$$

(iii) [2 marks]

Sketch an implementation of the Boolean expression $S = \overline{A + B} + BC$, using only AND, OR, and NOT logic gates.

(b) You are studying for an advanced computer architecture paper, and while trawling through your ENCE260 papers, you find Figures 1 and 2.

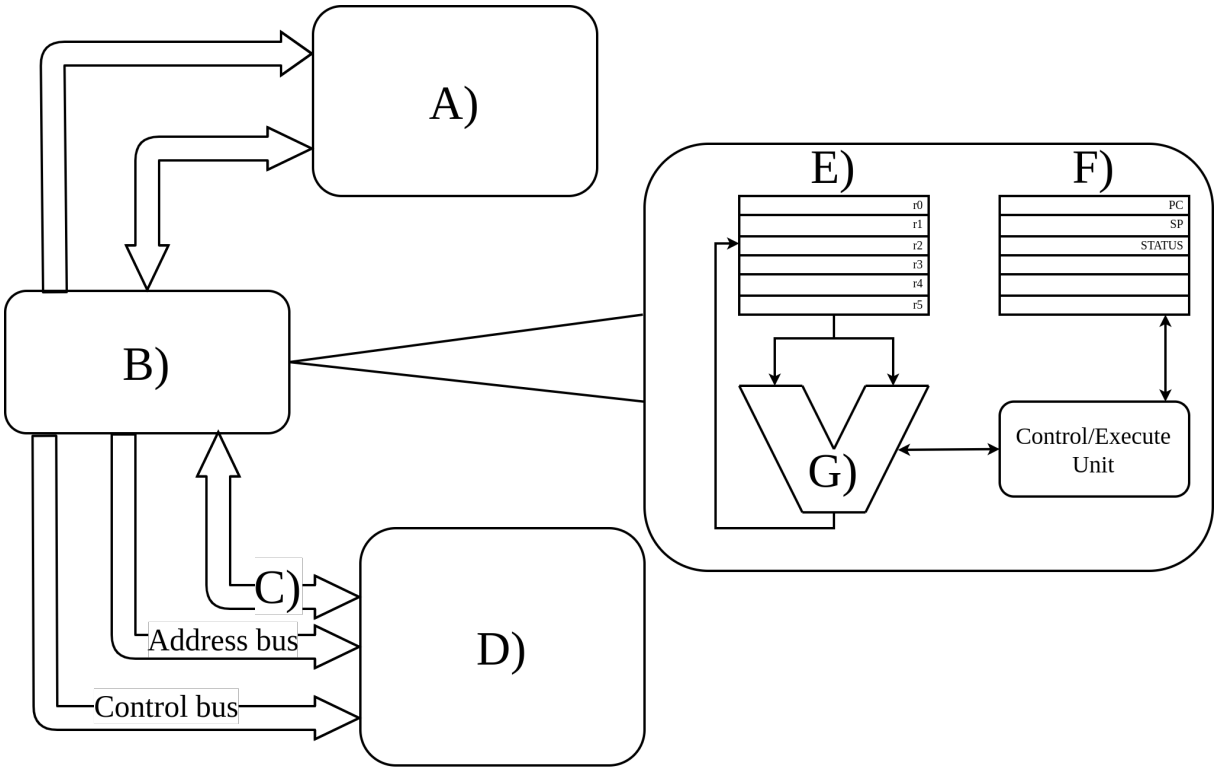


Figure 1: Block diagram showing a generic computer architecture.

(i) [4 marks]

Unfortunately, the diagram that you found does not have everything labelled. Give names for labels A) through G):

A)

B)

C)

D)

E)

F)

G)

(ii) [4 marks]

You have written in your notes that the picture in Figure 2 is a specific computer architecture, and therefore contains all of the same features as Figure 1.

On the diagram in Figure 2, label components A) through G).

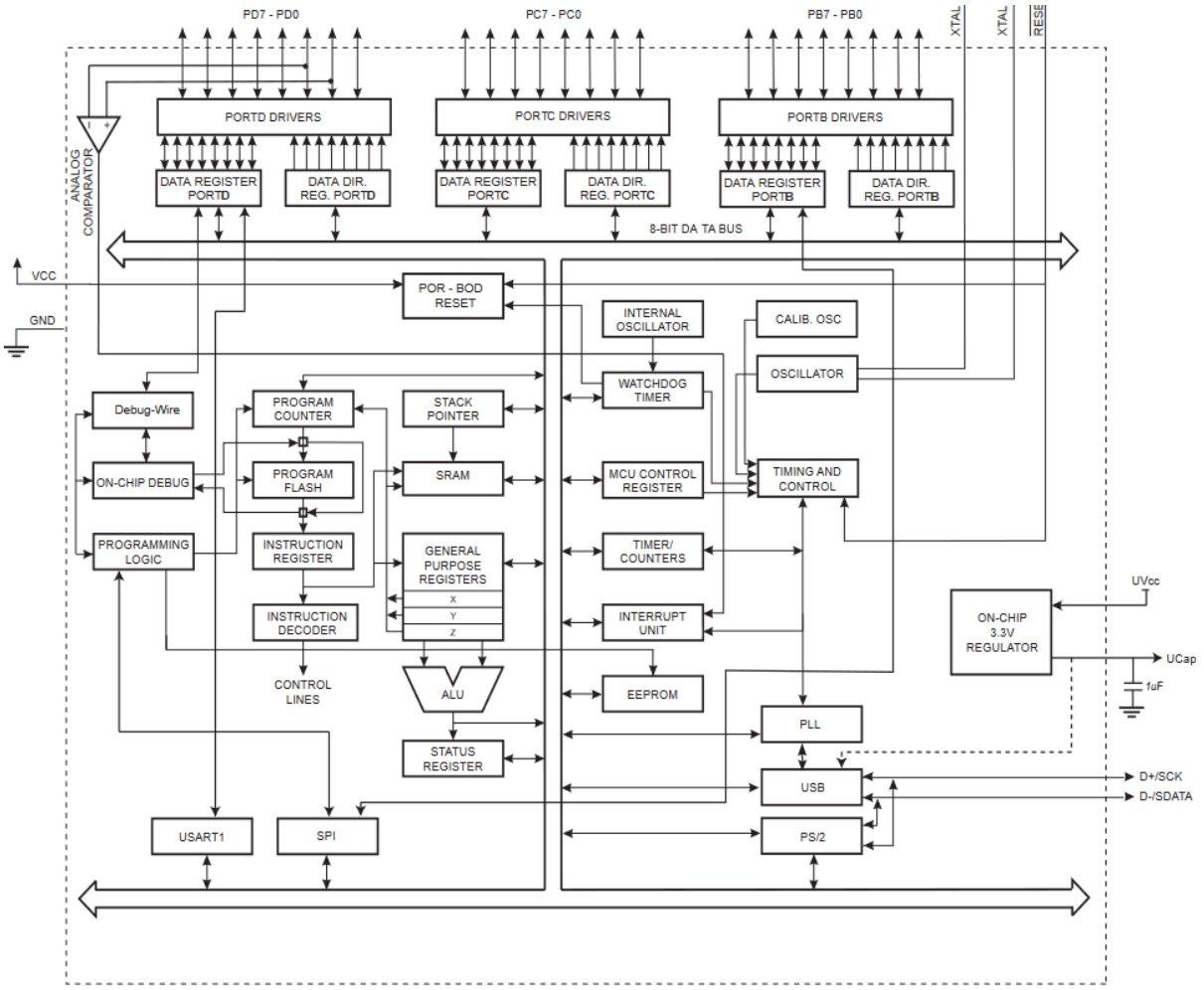


Figure 2: Block diagram of the ATMega32u2.

(iii) [2 marks]

In your notes you have written "*Harvard?? Von Neumann??*". State which architecture type the ATMega32u2 shown in Figure 2 is, with justification.

(c) Figure 3 is the truth table for an individual Full Adder circuit which performs addition on 1-bit numbers. The 4-bit Ripple-Carry Adder, shown in Figure 4, uses four Full Adder circuits to perform addition on 4-bit binary numbers.

Inputs			Outputs	
A	B	C_{in} (carry in)	S (sum)	C_{out} (carry out)
0	0	0	0	0
0	1	0	1	0
1	0	0	1	0
1	1	0	0	1
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	1

Figure 3: Truth table for a Full Adder.

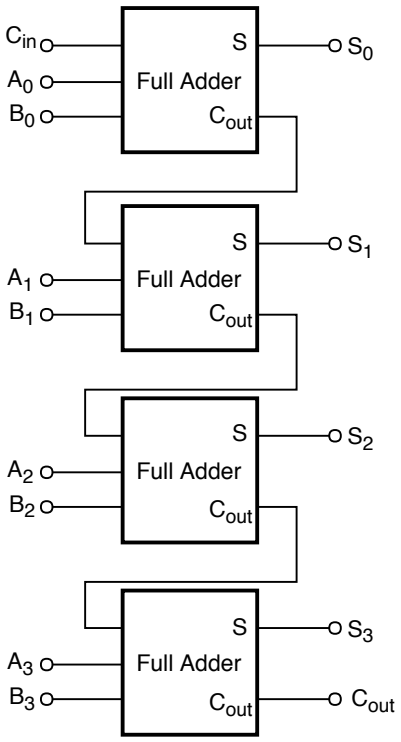


Figure 4: 4-bit Ripple-Carry Adder.

(i) [2 marks]

The Ripple-Carry Adder is to be used to sum the unsigned base-10 numbers A = 7 and B = 11. Write the 4-bit binary sequences, $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$, that represent the inputs to the Ripple-Carry Adder.

(ii) [3 marks]

Assuming that $C_{in} = 0$, perform the binary addition of $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ to find the values of the Sum, $S_3S_2S_1S_0$, and the C_{out} output.

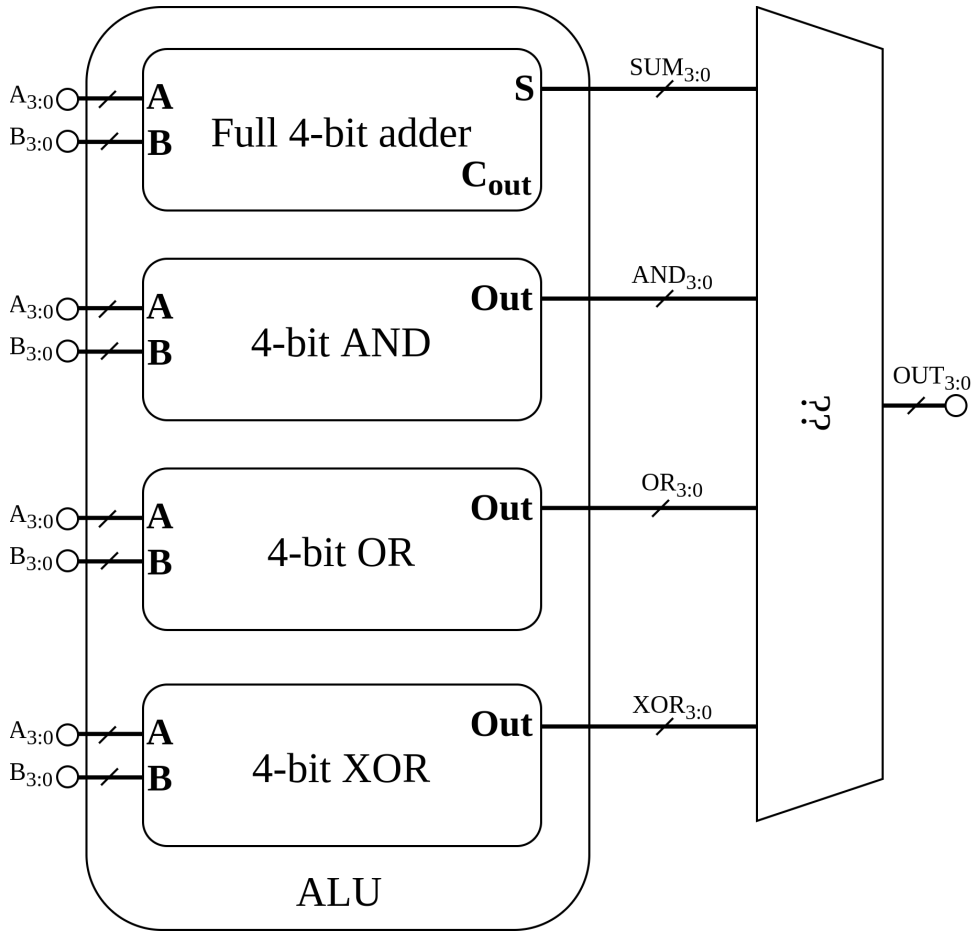
(iii) [2 marks]

The carry in, C_{in}, could permanently be set in hardware to $= 0$. With reference to two's complement number representation, state why it is useful to be able to set C_{in} electronically.

(iv) [1 mark]

Will all of the output signals, $S_3S_2S_1S_0$, change at *exactly* the same time?

Q2. **(a)** You are playing around with an ATMega32u2 that you have exposed, and you accidentally spill an extremely small drop of coffee onto it! You deduce that you have damaged some circuitry that selects the output of the arithmetic logic unit:



(i) **[1 mark]**

What is the name/type of circuit required which takes four inputs, and, on the basis of selector inputs, directly outputs one of the inputs?

On closer inspection, only one of the data lines is actually damaged. You go back to your old ENCE260 notes, where you find the following truth table, with the note “2:1":

Input			Output
S	I_1	I_2	O
0	0	X	0
0	1	X	1
1	X	0	0
1	X	1	1

(ii) [1 mark]

How many select lines are needed for a 4:1 circuit?

(iii) [1 mark]

The control/execute unit controls these select lines. The control/execute unit is an example of what type of combinational logic circuit?

You fix your ATMega32u2, and are now designing a system where it will interface with several buttons and LEDs, to act as inputs and outputs. Because you are not sure what is going to be connected where, programmable bidirectional GPIO pins are required.

Figure 12-2. General Digital I/O(1)

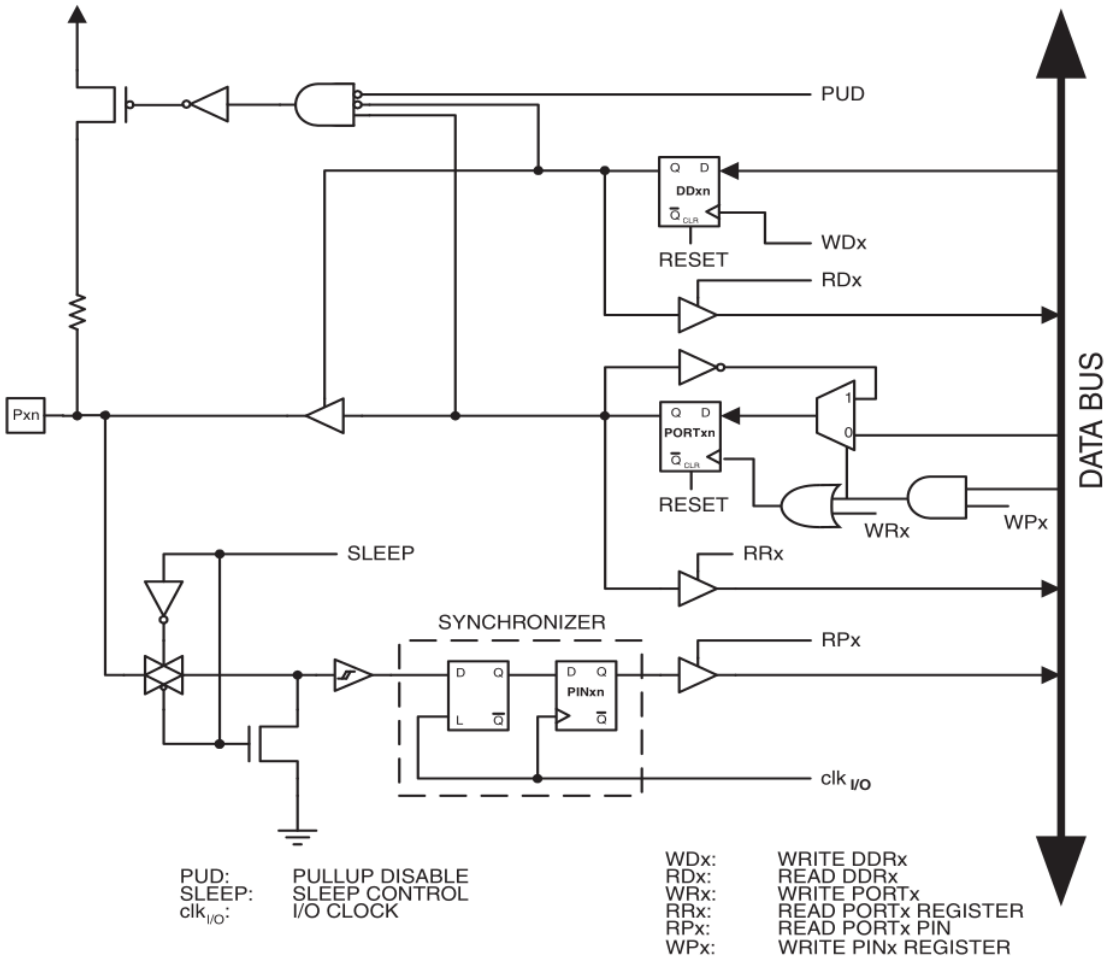


Figure 5: Diagram of the internal GPIO circuitry of the ATMega32u2.

(iv) [5 marks]

State the four possible operational states that enable bidirectional digital operation of the pin `Pxn`, and, with reference to Figure 5, state how this bidirectionality is achieved. In your answer, state the key registers that are used to manipulate the pin.

(b) You want to implement a system where an analogue signal is converted to a pulse-width modulated output.

(i) [4 marks]

Describe, with pictures and/or graphs, how approximation of an analogue signal is performed with use of an analogue comparator and a timer. Use your answer to justify why an analogue voltage can be determined with the formula:

$$V_{in} = V_{ref} \cdot \frac{T_{in}}{T_{ref}}$$

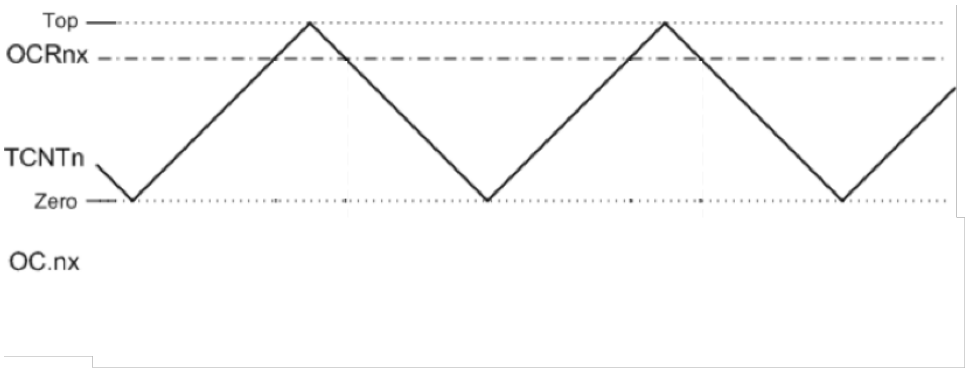


Figure 6: Diagram that illustrates how to generate a pulse width modulation output from a timer.

(ii) [3 marks]

To generate your PWM signal, someone asks which register or registers you need to modify to change the duty cycle. State which register you need to update, and, with reference to Figure 6, **justify your answer**.

(iii) [2 marks]

You know that if you connect you PWM output to an LED directly, you will need a resistor in series with the LED as well. State why this is, and suggest an alternative method to solve this same problem.

(c) (i) [4 marks]

Assuming you are using an AVR microcontroller, explain why the `brlt` opcode (summarised in Table 1) increments PC either by 1, like other opcodes, or some other value, and why this is useful.

Table 1: `brlt` opcode summary.

Syntax	Operands	Operation
`brlt k`	$-64 \leq k \leq +63$	Is result of CP, CPI, SUB, or SUBI < 0?
Control registers	Opcode	Flags
PC ← PC + 1 (false) PC ← PC + k + 1 (true)	111 00kk kkkk k100	None

(ii) [4 marks]

List four pieces of information that are typically stored on the stack in an AVR microcontroller.

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Q3. Embedded Systems 1

(a) An ATmega32u2 micrcocontroller is connected to four LEDs (L1 to L4) and two pushbutton switches (SW1 and SW2) as shown in Figure 7.

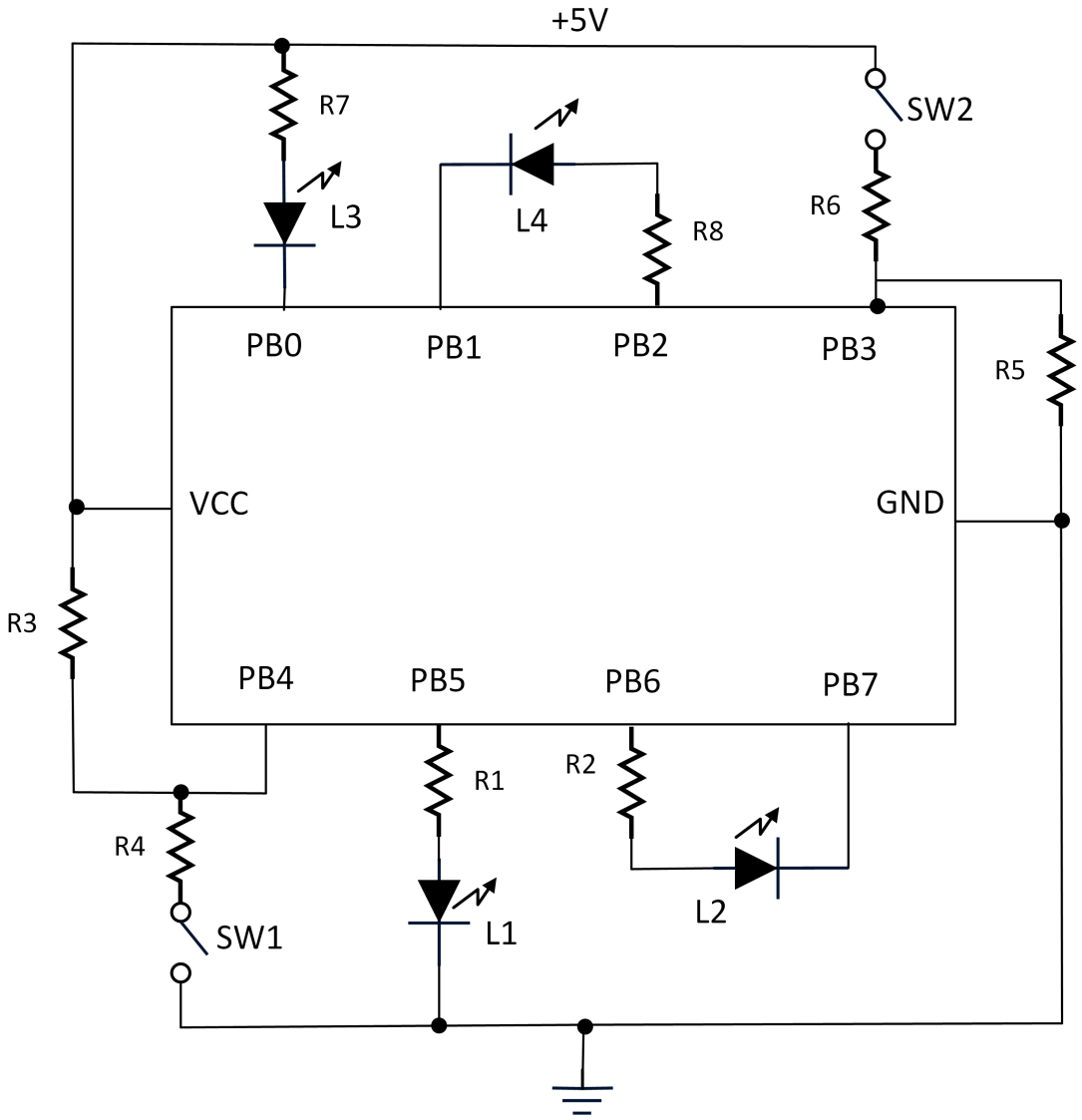


Figure 7: Schematic showing four LEDs and two pushbuttons connected to an ATmega32u2 microcontroller for Q3.

(i) **[2 marks]**

What is the purpose of resistor R6?

(ii) [2 marks]

Identify any pull-down resistors. What problem arises if a pull-down resistor is not present?

(iii) [2 marks]

What **hexadecimal** value should be stored in the Data Direction Register for Port B (DDRB)?

(iv) [2 marks]

What is the **smallest decimal** value that can be stored in the `PORTB` register so that only the L1 and L2 LEDS are on?

(v) [2 marks]

Write a C statement that will toggle L1 without affecting the other outputs.

(vi) [2 marks]

Write an `if` conditional statement in C that evaluates to `True` when the pushbuttons SW1 and SW2 are simultaneously pushed.

Question 3 continues on p22

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(b) Rocky and Skye programmed `tictactoe` as their game on the UCFK4. The makefile for this code is shown in Listing 1. Both Rocky and Skye have this version of `tictactoe` in their local repositories and in the remote repository.

Listing 1: Makefile for Question 3(b).

```
CC = avr-gcc
CFLAGS = -mmcu=atmega32u2 -Os -Wall
OBJCOPY = avr-objcopy
SIZE = avr-size
DEL = rm

all: tictactoe.out

tictactoe.o: tictactoe.c ../drivers/avr/system.h \
        ../tictactoe.h
    $(CC) -c $(CFLAGS) $< -o $@

system.o: ../drivers/avr/system.c ../drivers/avr/system.h
    $(CC) -c $(CFLAGS) $< -o $@

pio.o: ../drivers/avr/pio.c ../drivers/avr/pio.h \
        ../drivers/avr/system.h
    $(CC) -c $(CFLAGS) $< -o $@

led.o: ../drivers/led.c ../drivers/avr/pio.h \
        ../drivers/avr/system.h ../drivers/led.h
    $(CC) -c $(CFLAGS) $< -o $@

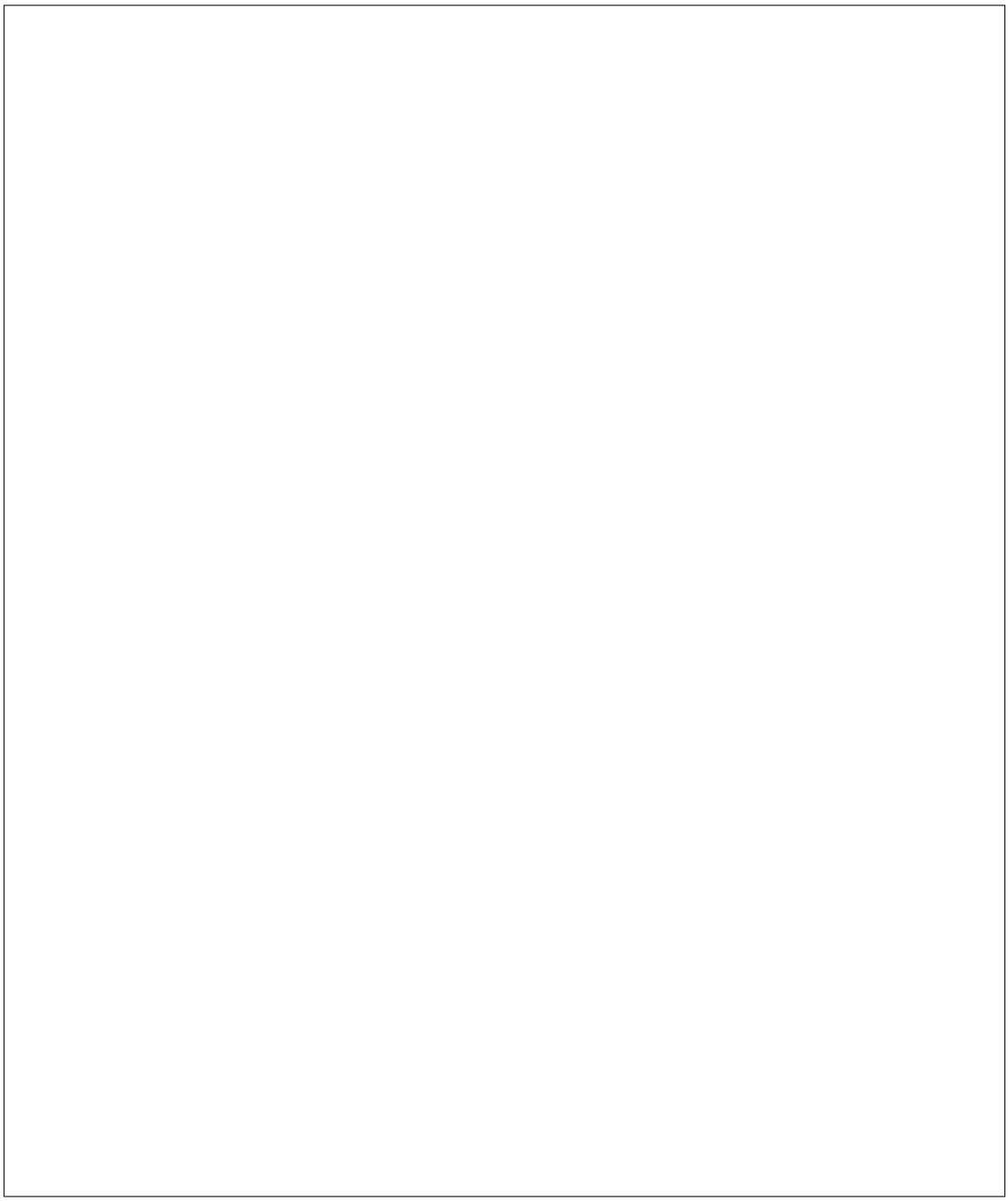
tictactoe.out: tictactoe.o pio.o system.o led.o
    $(CC) $(CFLAGS) $^ -o $@ -lm
    $(SIZE) $@

.PHONY: clean
clean:
    -$(DEL) *.o *.out *.hex

.PHONY: program
program: tictactoe.out
    $(OBJCOPY) -O ihex tictactoe.out tictactoe.hex
    dfu-programmer atmega32u2 erase; dfu-programmer \
        atmega32u2 flash tictactoe.hex; \
            dfu-programmer atmega32u2 start
```

(i) [3 marks]

In the box below, draw a dependency graph of the files used to build the executable file. Note, you are not required to show the programming tools used in the build. For clarity and ease in drawing your graph, you may ignore all dependencies on `system.h`.



(ii) [2 marks]

The makefile in Listing 1 on Page 22 contains a number of automatic variables as indicated by the $ symbol. Write out Line 25 of Listing 1 in full.

(iii) [2 marks]

Of the file types used to produce the `tictactoe` program, which should be included in a version control system such as git?

(iv) [2 marks]

What will the command `make program` do here?

(v) [2 marks]

Rocky decides to add another module `button.c` and its header file `button.h` to the final build. `button.c` is also dependent on the `pio` module. In the answer box below, write or rewrite the line or lines in the makefile required to compile and link this new module.

(vi) [1 mark]

Rocky makes your suggested changes and updates the remote repository. Use the following box for the git command(s) that Skye needs to write so that her local repository includes these changes from Q3(b)v.

(vii) [1 mark]

Skye notices a bug in this new version and wishes to remove Rocky's changes. Use the following box to write the git command(s) to modify Skye's local repository back to the previous version.

Q4. Embedded Systems 2

(a) Data is transmitted serially from one UART to another. Assume the UARTs have been configured for 7-bit data, the LSB is transmitted first, the idle state is high, and there are one stop bit, one start bit, and one odd parity bit. Answer the following:

(i) [4 marks]

Sketch the waveform that should be transmitted from the UART to transmit the ASCII character `b`. An ASCII table is shown in Figure 8.

(ii) [3 marks]

If the Baud rate is 14400 Baud, how long will it take to transmit 256 characters?

0	NUL	32		64	@	96	‘
1	^A	33	!	65	A	97	a
2	STX	34	"	66	B	98	b
3	ETX	35	#	67	C	99	c
4	^D	36	$	68	D	100	d
5	^E	37	%	69	E	101	e
6	^F	38	&	70	F	102	f
7	BELL	39	’	71	G	103	g
8	BS	40	(	72	H	104	h
9	TAB	41	)	73	I	105	i
10	LF	42	*	74	J	106	z
11	VT	43	+	75	K	107	k
12	FF	44	,	76	L	108	l
13	CR	45	-	77	M	109	m
14	^N	46	.	78	N	110	n
15	^O	47	/	79	O	111	o
16	^P	48	0	80	P	112	p
17	XON	49	1	81	Q	113	q
18	^R	50	2	82	R	114	r
19	XOFF	51	3	83	S	115	s
20	^T	52	4	84	T	116	t
21	^U	53	5	85	U	117	u
22	^V	54	6	86	V	118	v
23	^W	55	7	87	W	119	w
24	^X	56	8	88	X	120	x
25	^Y	57	9	89	Y	121	y
26	^Z	58	:	90	Z	122	z
27	ESC	59	;	91	[	123	{
28	FS	60	<	92	\	124	|
29	GS	61	=	93	]	125	}
30	RS	62	>	94	^	126	~
31	US	63	?	95	_	127	DEL

Figure 8: The ASCII character table for Q4(a).

(b) Chase and Marshall have written the C program `blinking.c` in Listing 2 for the ATmega32u2 microcontroller. The TCCR0B configuration register is shown in Figure 9, and the TCNT0 register in Figure 10. LED1 is in an active low configuration. Listing 2 has been committed by Chase to their group's git repository. Given a CPU frequency of $4\,\mathrm{MHz}$, answer the following.

Listing 2: Source listing for Q4(b)

```
1  #include "pio.h"
2  #include "system.h"
3
4  void pacer (uint8_t ticks) {
5      while (TCNT0 < ticks)
6          continue;
7      TCNT0 = 0;
8  }
9
10 void blinking(uint8_t alpha, uint8_t beta, uint8_t gamma)
11 {
12     TCCR0B = 0x05;
13     TCNT0 = 0;
14
15     while (1)    {
16
17         pio_output_low(LED1);
18         pacer(alpha+beta);
19         pio_output_high(LED1);
20         pacer(gamma+gamma*(alpha/beta));
21     }
22 }
23
24 int main (void)
25 {
26     system_init ();
27     pio_config_set(LED1, PIO_OUTPUT_HIGH);
28     blinking(0x64,0xC8,0x50);
29
30     return 0;
31 }
```

Bit	7	6	5	4	3	2	1	0	
0x25 (0x45)	FOC0A	FOC0B	–	–	WGM02	CS02	CS01	CS00	TCCR0B
Read/Write	W	W	R	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 15-9. Clock Select Bit Description

CS02	CS01	CS00	Description
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	$clk_{I/O}$/(No prescaling)
0	1	0	$clk_{I/O}$/8 (From prescaler)
0	1	1	$clk_{I/O}$/64 (From prescaler)
1	0	0	$clk_{I/O}$/256 (From prescaler)
1	0	1	$clk_{I/O}$/1024 (From prescaler)
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

Figure 9: TCCR0B register bits for Q4(b).

TCNT0 – Timer/Counter Register

Bit	7	6	5	4	3	2	1	0	
0x26 (0x46)	TCNT0[7:0]								TCNT0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Figure 10: TCNT0 register bits for Q4(b).

(i) [1 mark]

What is the purpose of Line 6 in Listing 2?

(ii) [2 marks]

What is the timing resolution of the timer in Listing 2?

(iii) [5 marks]

What is the duty cycle of the blinking LED in Listing 2?

(iv) [2 marks]

Marshall peer reviews his partner Chase's code and finds several examples of poor programming practice (i.e. not in the calculations from lines 17-20, but in how the rest of the code is written). Suggest two examples of poor programming practice from Chase.

(v) [2 marks]

Use the answer box below to add appropriate lines of code and/or rewrite others to show how Chase can correct **one** example of poor programming practice identified in Q4(b)iv.

(vi) [2 marks]

Chase implements the changes you suggest. In the following box, write the git command(s) to ensure this file will be up-to-date in their remote repository.

(c) [4 marks]

Describe two different approaches for working together with a colleague to write a software program, such as a game on the UCFK4. (Note that all approaches should use a version control system such as git, so you don't need to include this in your answer). Give one advantage and one disadvantage for one of these approaches.

END OF QUESTIONS

Extra answer space (optional)

Extra answer space (optional)

END OF EXAM